DGMS Examination 2019 — Question 57 Detailed Solution

Junior Scientific Officer (Geology) | Structural Geology & Boudinage Mechanics

QUESTION 57

The separation of a competent layer due to layer-parallel extension is incomplete and shows a narrowing of the segments. This structure is termed as:

(a) Pinch-and-swell structure CORRECT ANSWER	(b) Chocolate-tablet structure
(c) Extension fracture boudinage	(d) Tension gash

1. Executive Summary & Answer Verification

The correct option is **(a) Pinch-and-swell structure**.

In structural geology (codified by Ramsay, 1967; Ramsay & Huber, 1987; Haakon Fossen, 2016):

- When a stiff, competent layer (such as a sandstone bed, quartz vein, or amphibolite dike) embedded in a more ductile matrix (such as shale, schist, or marble) undergoes **layer-parallel extension**:
  - If the competent layer deforms plastically/ductilely by localized tensile thinning (the process of *necking*) without complete rupture, the layer remains continuous. It forms alternating thickened segments ("swells") and constricted, narrowed segments ("pinches").
  - This morphological structure where separation is **incomplete** and segments remain connected across narrowed necks is universally termed **Pinch-and-swell structure**.
- When the extension proceeds further until brittle tensile fractures or shear fractures cut completely through the narrowed necks, the segments become isolated sausages called **boudins** (boudinage).

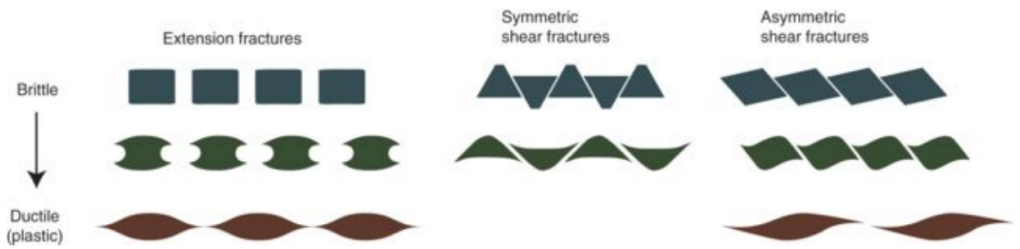


Figure 15.1 The geometry of boudins is largely controlled by whether boudins are separated by extension or shear fractures, and the influence of plastic versus brittle deformation mechanisms. Asymmetric boudins may indicate non-coaxial deformation.

There are also examples of regularly spaced areas of thinning in

Figure 57.1: *Geometries of extended competent layers under layer-parallel extension: (Top) Extension fractures and shear fractures leading to discrete boudins; (Bottom)* ***Pinch-and-swell structure*** *where plastic necking produces narrowing of segments with incomplete separation. (Haakon Fossen, Structural Geology, 2nd ed., Chapter 15, Figure 15.1, p. 336).*

2. Rheology & Mechanics of Layer Extension

2.1 The Competence Contrast & Plastic Necking

The physical behavior of an extended layer depends on the viscosity/ductility contrast between the layer (μ_{layer}) and matrix (μ_{matrix}):

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m = \frac{\mu_{\text{layer}}}{\mu_{\text{matrix}}} > 1
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1. **Ductile Necking (Pinch-and-Swell):** Under high metamorphic temperatures and pressures (greenschist to amphibolite facies), both the layer and matrix deform by crystal plastic dislocation creep. Because the layer is more viscous ($m > 1$), small initial thickness perturbations amplify under extension into regular wavelength necking zones. Because plastic strain accommodates the extension, no physical gap forms, maintaining an unbroken, undulating layer of pinches and swells.
2. **Brittle Rupture (Fracture Boudinage):** At lower temperatures or higher strain rates, the competent layer exceeds its tensile strength before plastic necking can fully develop. Cross-cutting extension joints form perpendicular to the layering, severing the layer into separate boudins with mineral-filled scarps (veins) in the inter-boudin gaps.

2.2 Evaluation of Other Structural Terms

- **(b) Chocolate-Tablet Structure:** Occurs during *biaxial extension* (two orthogonal extension directions within the layer plane: $\lambda_1 > 1, \lambda_2 > 1$). Two perpendicular sets of boudin necks develop simultaneously, dicing the competent layer into rectangular, tablet-like blocks resembling a segmented chocolate bar. Here, separation is complete in two dimensions, not an incomplete single-direction pinch-and-swell.
- **(c) Extension Fracture Boudinage:** Refers specifically to boudinage formed by complete brittle tensile fracturing through the layer, leaving distinct rectangular blocks separated by open or vein-filled fractures.

- **(d) Tension Gash:** A lenticular, sigmoidal, or planar brittle tensile fissure formed perpendicular to the maximum local tensile stress (σ_3), typically occurring in en-echelon arrays within shear zones. It is not an undulating, thinned layer.

3. Systematic Comparison of Extensional Layer Structures

Structure	Continuity of Layer	Dominant Mechanism	Geometric Appearance
Pinch-and-Swell	**Continuous (Incomplete separation)**	**Plastic necking / ductile stretching**	**Alternating swollen pods and narrowed necks**
Extension Boudinage	Discontinuous (Complete separation)	Brittle tensile fracturing	Rectangular blocks with square corners separated by veins
Shear Boudinage	Discontinuous (Complete separation)	Conjugate shear fracture slipping	Rhomboidal or lozenge-shaped asymmetric blocks
Chocolate-Tablet	Discontinuous (2D separation)	Biaxial 3D extension in layer plane	Rectangular tablets segmented by two orthogonal neck sets

Textbook Authority

Haakon Fossen, *Structural Geology* (2nd ed., Chapter 15: Boudinage, p. 336):
"There are also examples of regularly spaced areas of thinning in many extended competent layers without the separation into isolated fragments or boudins. Such structures are called pinch-and-swell structures and the process is called necking."
John G. Ramsay & Martin I. Huber, *The Techniques of Modern Structural Geology, Volume 2: Folds and Fractures*:
"Where the competence contrast is moderate and the materials deform plastically, extension produces necking without actual rupture, giving pinch-and-swell structure. Only when extension exceeds the plastic limit do individual boudins become completely separated."